



Automated Precision Experimentation Laboratories

ILR05

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1 Individual Progress

During the Progress Review 4 period, my primary technical focus was on advancing the planning subsystem toward full integration with both the perception and manipulation stacks. The work this period built directly on the MoveIt and IsaacSim foundation established in PR3, shifting emphasis from simulation validation toward real sensor data and cross-subsystem integration. The planning goals for this period were partially achieved, with meaningful progress made on live point cloud ingestion and AprilTag-based target localization, while full gripper and behavior tree integration remains ongoing. These efforts represent a meaningful step toward closing the loop between what the robot sees and what it plans to do.

1.1 Live point cloud integration

A significant milestone this period was loading point cloud data from the live ZED camera feed directly into the planning stack, replacing the dummy point clouds used in prior simulation testing. This data is consumed by the MoveIt collision environment as a voxel occupancy map, enabling the planner to avoid obstacles observed in real time rather than relying on a static scene description. Bringing live sensor data into the collision world is a prerequisite for safe autonomous operation in the lab environment, where the positions of equipment such as the liquid handler, shaker module, and OT2 deck are not known to the planner a priori and must be inferred from depth sensing.

A persistent challenge encountered during this process was that the point cloud occasionally classified the robot arm itself as a collision object, causing the planner to treat its own links as obstacles and reject otherwise valid trajectories. This happens because the arm appears in the camera's field of view and its depth returns are included in the raw point cloud before any filtering is applied. Resolving this requires careful configuration of the self-filter and the robot's published TF frames so that MoveIt can correctly exclude the arm's own geometry from the occupancy grid. A robot-aware filter that projects the current joint configuration into the camera frame and masks those pixels before voxelization is the approach being pursued.



Figure 1: Point cloud in RViz

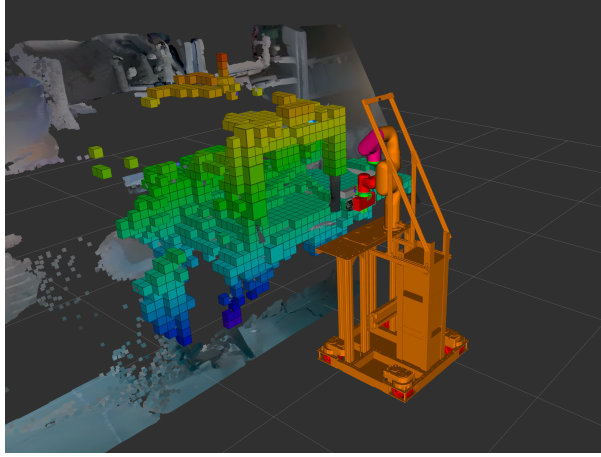


Figure 2: Voxel maps in RViz

1.2 AprilTag-Based Target Localization

Path planning to a designated target location using AprilTag detection was successfully demonstrated this period. The system uses an AprilTag placed at the OT2 slot as a localization reference, and the planner generates a collision-free trajectory to bring the end-effector to that position. Detection of the tag provides a 6-DOF pose estimate in the camera frame, which is then transformed into the robot's base frame using the known camera-to-base extrinsic calibration, and passed to MoveIt as the Cartesian goal pose for the end-effector. This represents the first instance of the planning stack consuming externally perceived spatial information to determine a goal pose, rather than relying on a hard-coded target. It establishes the integration pattern that will be used once the object detection pipeline outputs full 3D coordinates for wellplate targets. The same transform chain, from camera frame detection through extrinsic calibration to robot base frame goal, will be reused when switching from AprilTag localization to YOLO-based wellplate detection, making this a reusable and well-validated pathway into the planner.

<https://www.youtube.com/watch?v=jBP-e2X0ZQA>

2 Challenges

The orientation constraint requiring the end-effector to remain strictly downward-facing throughout all trajectories continues to restrict the available configuration space significantly. This constraint is necessary to prevent the wellplate contents from spilling during transport, but it eliminates large portions of the joint space that would otherwise be available to the stochastic sampler. RRTConnect's sampling-based search struggles in these narrow corridors of valid configurations, producing occasional planning timeouts on paths that are geometrically straightforward. This remains an active area of investigation and deterministic planner alternatives, such as STOMP or a Cartesian interpolation fallback for constrained segments, are being evaluated to improve reliability and consistency across repeated planning calls. The point cloud filtering

issue where the arm is treated as a collision object is the most pressing technical problem on the planning side. Until this is resolved, the planner will occasionally fail to find valid paths even when the physical environment is clear, which reduces overall system reliability for end-to-end testing and makes it difficult to distinguish genuine collision avoidance failures from self-occlusion artifacts.

3 Teamwork

- **Arnav** worked on visual servoing for grasping the wellplate, successfully demonstrating the full grasp and transfer sequence on the physical xArm6.
- **Juan** worked on getting the liquid handler to dispense liquid correctly and helped optimize the camera stream pipeline across the Jetson units.
- **Farnaz** focused on the perception pipeline, working on fine tuning the YOLO models for our use case.

4 Plans

The immediate priority is completing the integration between the planning stack and the perception output. Once object detection begins publishing 3D target coordinates for the wellplate, those will be fed directly into the goal pose for the planner, allowing end-to-end verification of the perception and planning pipeline together. The AprilTag localization work this period has already validated the transform chain needed to do this, so the transition to YOLO-based goal poses should be straightforward once the detection output is available in a stable ROS topic.

Visual servoing is complete on the manipulation side, meaning planning and manipulation integration can be tested as the next concrete step without waiting on any outstanding dependencies. The plan is to command the arm to a pre-grasp pose above the wellplate via MoveIt, hand off to the visual servo controller for final approach, and verify that the two subsystems transition cleanly without requiring manual intervention between them. Parallel to this, the planning stack needs to be connected to the behavior tree so that grasp goals can be issued autonomously rather than triggered manually from the command line. Once that interface is established, the team will begin dry runs of the full SVD workflow, tuning planner parameters and point cloud filtering as issues surface during those runs.